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**Roll No** .....

# EX-8302 (GS)

## B.E. VIII Semester Examination, June 2020

## Grading System (GS)

## Advanced Communication Systems

### (Elective - III)

***Time : Three Hours***

***Maximum Marks : 70***

***Note:*** i) Attempt any Five questions.
ii) All questions carry equal marks.

1. Draw block diagram of BPSK communication system incorporating a spread spectrum technique.

2. a) Explain the basic concept of spread spectrum and discuss spread spectrum with CDMA.
   b) List the various frequency bands used in satellite communication. Describe the characteristics of a geostationary orbit with its uses.

3. a) A 1000 inlet and 1000 outlet digital switch is to be built using TSI. Determine the size of the control and data memories and the speeds with which the memories have to be accessed.
   b) How many subscribers can be supported in bidirectional PAM switching bus if the pulse width of the PAM sample is 125 ns?

4. Define:
   i) EIRP
   ii) Up-down converter

5. Draw satellite orbital patterns and define the following terms:
   i) Apogee ii) Major axis
   iii) Perigee iv) Minor axis

6. What characteristic can be singled out in the CDMA standard that distinguishes it from other adopted systems? Which are the negative aspects of the CDMA standard.

7. Illustrate on the multiple access methods used in satellite communication.

   OR

   Derive the expression of effective isotropic radiated power of satellite system.

8. Write Short Notes on (any two)
   a) FAX b) FHSS
   c) Tracking and Servo system d) XDSL

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